Speech to Speech Translation Systems

Benchmarking and Implementation

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Outline

Introduction

A usual S2S (speech-to-speech) pipeline consists of 3 main components - ASR (Automatic Speech Recognition), MT (Machine Translation) and TTS (Text-to-Speech). The ASR module converts the input speech into text, the MT module translates the text from source to target language and the TTS module converts the translated text back into speech.

Previously, each stage was implemented separately and trained over different datasets. However, this leads to error propagation and information loss across the stages. Current state-of-the-art systems implement the complete pipeline as a unified system. This allows for better retention of prosodic and acoustic features across the stages, leading to more natural and accurate translations.

ASR: Overview

The ASR module of a S2S pipeline first converts the input speech into text. It involves several key components:

- **Audio Preprocessing:** Noise reduction and feature extraction using MFCCs, Mel spectrograms etc.
- **Encoder:** Responsible for capturing temporal dependencies and contextual information of the input audio using the extracted features. Typically implemented using RNNs, CNNs, or Transformers.
- **Decoder:** Generates the output text sequence based on the encoder reprensentation. Current state-of-the-art models often use attention mechanisms to focus on relevant parts of the input during decoding.

ASR Benchmarking: Setup

We run our benchmarking on the LibriSpeech dataset from which 100 samples of english speech (around 8 minutes of audio). The audio signals are resampled to 16 kHz and normalised to be all lowercase and without punctuation. The following models are evaluated:

- **OpenAI Whisper - Tiny:** Around 39M parameters
- **OpenAI Whisper - Base:** Around 74M parameters
- **OpenAI Whisper - Small:** Around 244M parameters
- **Distill Whisper - Small:** Around 166M parameters
- **OpenAI Whisper - Medium:** Around 394M parameters

We use word error rate (WER), character error rate (CER), sentence error rate (SER) and real-time factor (RTF) as our evaluation metrics.

ASR Benchmarking: WER Results

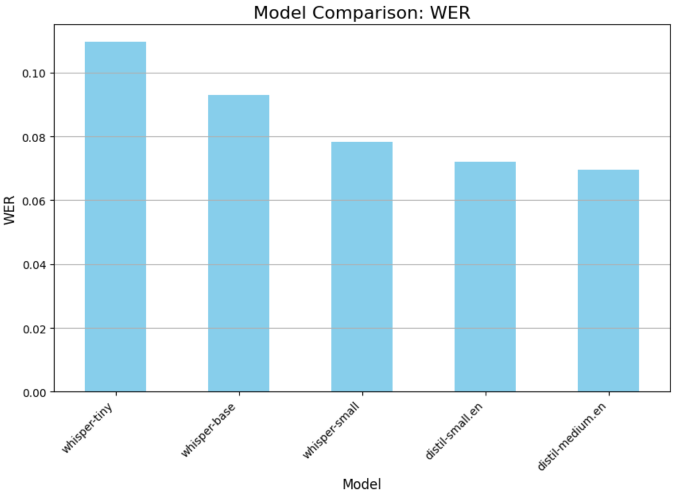


Figure: WER results for different ASR models

ASR Benchmarking: CER Results

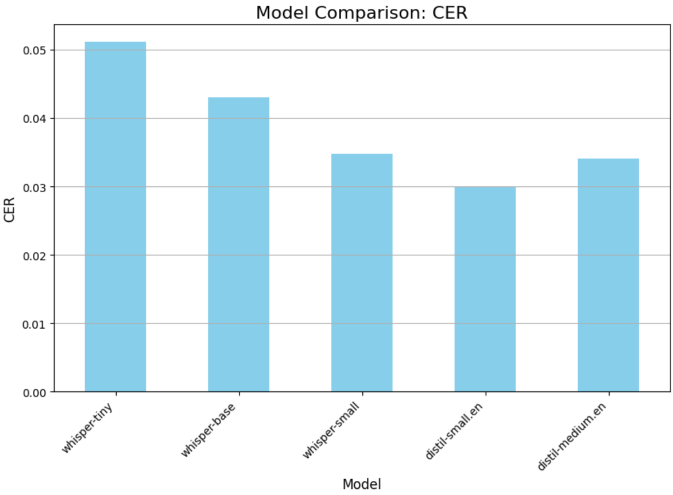


Figure: CER results for different ASR models

ASR Benchmarking: SER Results

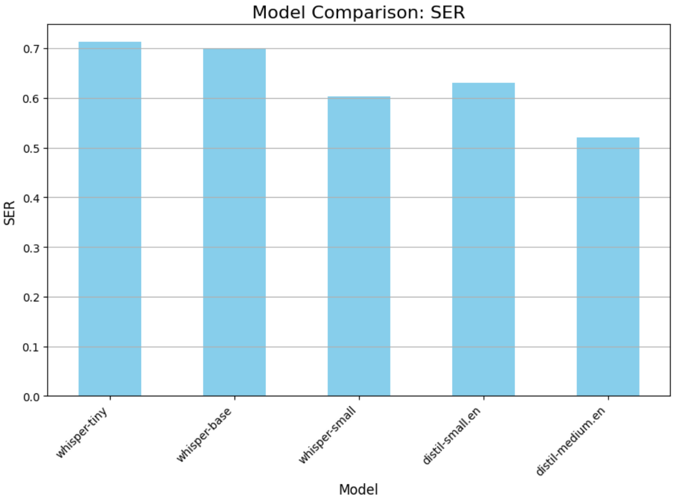


Figure: SER results for different ASR models

ASR Benchmarking: RTF Results

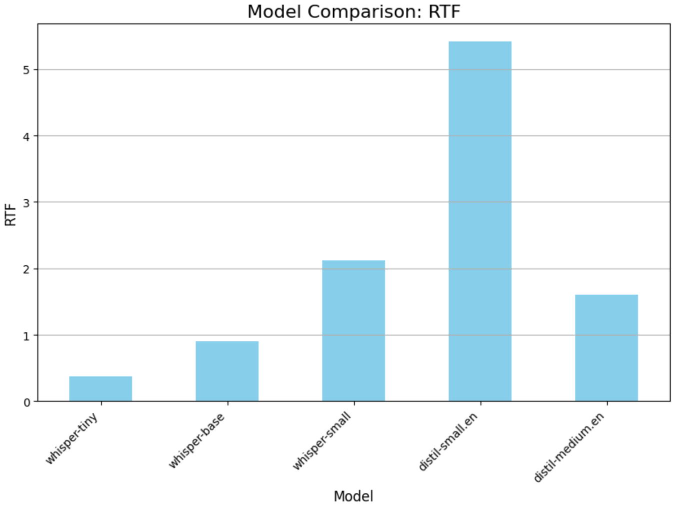


Figure: RTF results for different ASR models

ASR Benchmarking: Inferences

- Clear improvement in WER with increasing model size, with the medium model achieving the lowest WER of close to 0.07. The Distill small model performs better than OpenAI base model despite having fewer parameters.
- CER shows similar trends however the Distill medium model performs slightly worse.
- All models perform similary in terms of SER, with the Distill medium model performing the best.
- Distill small has the worst RTF. This is an anomalous result as the Distill medium performs much better with the same running conditions.

MT: Overview

- Translates text from source to target language
- Evolution: Rule-based $\rightarrow$ Statistical $\rightarrow$ Neural
- Transformer architecture revolution
- Quality assessment and evaluation

MT: Key Approaches

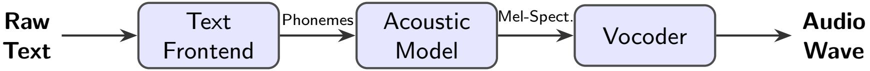
- Neural Machine Translation (NMT)
- Transfer learning and multilingual models
- Low-resource language handling
- Context-aware translation

What is Text-to-Speech (TTS)?

Text-to-Speech (TTS) is the artificial synthesis of human-sounding speech from raw text.

- **Core Objectives:**
  - *Intelligibility:* Clear, understandable pronunciation of words.
  - *Naturalness:* Accurate prosody, intonation, rhythm, and emotion.
- **Evolution of the Technology:**
  - **Concatenative (1990s-2000s):** Splicing together tiny fragments of pre-recorded human speech. Highly intelligible but sounded robotic and rigid.
  - **Parametric (2000s-2010s):** Statistical models (like HMMs) generating audio based on parameters. Smoother, but sounded muffled or "buzzy".
  - **Neural (Present):** Deep learning models predicting high-fidelity audio waves directly from text, achieving near-human naturalness.

The Standard Neural TTS Pipeline



- **Text Frontend:** Cleans and normalizes text (e.g., expanding "180" to "one hundred eighty") and performs Grapheme-to-Phoneme (G2P) conversion to determine pronunciation.
- **Acoustic Model:** Maps phonemes to acoustic representations, predicting pitch, energy, and duration. *(Examples: Tacotron 2, FastSpeech)*
- **Vocoder:** Reconstructs the actual time-domain audio waveform from the highly compressed Mel-spectrogram. *(Examples: WaveNet, HiFi-GAN)*

Systems Under Evaluation

MMS-TTS

- **Architecture:** VITS-based (End-to-End).
- **Primary Strength:** Highly lightweight and efficient; state-of-the-art for low-resource languages.
- **Limitation:** Generally single-speaker per model with fixed prosody.

Parler-TTS

- **Architecture:** Autoregressive Transformer.
- **Primary Strength:** Granular control over emotion, tone, pacing, and background noise via text descriptions.
- **Limitation:** Higher inference latency due to autoregressive generation.

XTTS-v2

- **Architecture:** GPT-style + Mel/HiFi-GAN.
- **Primary Strength:** Requires only ~3 seconds of reference audio; supports cross-lingual voice cloning.
- **Limitation:** High resource consumption compared to non-autoregressive models.

Evaluation Metrics for TTS Models

Intelligibility & Efficiency

- **Word Error Rate (WER)**
  What it is: Transcription accuracy of the audio using an ASR system. Lower is better.
- **Character Error Rate (CER)**
  What it is: Similar to WER, but captures finer phonetic/spelling errors. Lower is better.
- **Real-Time Factor (RTF)**
  What it is: Processing time divided by generated audio length. Lower is better.

Prosody & Expressiveness

- **Pitch Variance (F_0 Std Dev)**
  What it is: Range of intonation.
- **Energy Dynamics (RMS Std)**
  What it is: Variations in volume/loudness.
- **Speaking Rate**
  What it is: Speed (Syllables/words per sec).
- **Pause Ratio (Onset/s)**
  What it is: Frequency of breath/silence pauses.

TTS: Benchmarking Results

Model	Language	Latency (ms)	RTF	Throughput (cps)
MMS-TTS	English	2621.382	0.439	31.711
MMS-TTS	Hindi	2451.2891	0.432	27.49
Parler-TTS	English	68026.311	11.264	1.281
XTTS-v2	English	26182.9	3.438	3.491
XTTS-v2	Hindi	23687.88	3.204	2.718

Table: Benchmarking results for TTS models across Performance Metrics.

TTS: Benchmarking Results

Model	Language	WER	CER
MMS-TTS	English	34.892	17.968
MMS-TTS	Hindi	-	-
Parler-TTS	English	27.57	19.854
XTTS-v2	English	20.465	6.825
XTTS-v2	Hindi	-	-

Table: Benchmarking results for TTS models across Performance Metrics (Contd.).

**WER/CER for Hindi were turning out to be extremely high due to ASR model's poor performance on Hindi audio, so we have omitted those values to avoid skewing the analysis.*

TTS: Benchmarking Results

Model	Language	Pitch (Mean)	Pitch (Std)	Pitch Range
MMS-TTS	English	99.978	19.158	81.388
MMS-TTS	Hindi	148.216	37.359	176.392
Parler-TTS	English	233.055	54.52	253.809
XTTS-v2	English	141.029	29.674	136.078
XTTS-v2	Hindi	139.625	16.915	76.358

Table: Benchmarking results for TTS models across Prosodic Metrics (Contd.).

TTS: Benchmarking Results

Model	Language	Speaking Rate	Energy Dynamics	Pause Ratio
MMS-TTS	English	6.788	0.067	0.183
MMS-TTS	Hindi	5.808	0.093	0.246
Parler-TTS	English	5.35	0.082	0.153
XTTS-v2	English	4.284	0.099	0.212
XTTS-v2	Hindi	4.268	0.1002	0.186

Table: Benchmarking results for TTS models across Prosodic Metrics.

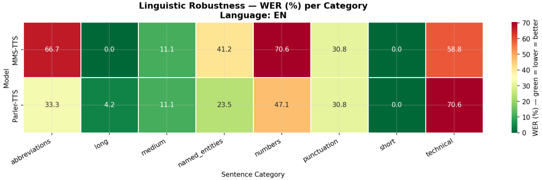


Figure: WER Heatmap for English for MMS-TTS and Parler-TTS for different types of sentences.

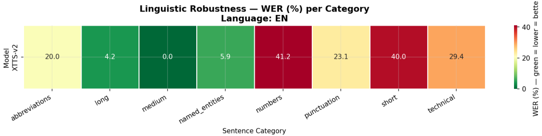


Figure: WER Heatmap for English for XTTS-v2 for different types of sentences.

End-to-End Pipeline

We also implemented a full S2S pipeline using few of the models that we benchmarked here and compared them against the existing SeamlessM4T-V2 (2.3B parameters) model. The models that we used for the pipeline were:

- **ASR:** OpenAI Whisper - Base (74M parameters)
- **MT:** Helsinki-NLP Opus-MT (English to Hindi) (77M parameters)
- **TTS:** MMS-TTS by Meta AI (559M parameters)

S2S pipeline: Scoring Metrics

We use the following scoring metrics for the end-to-end pipeline:

- **METEOR:** Measure n-gram precision and recall while accounting for synonyms and stemming.
- **chrF++:** Extends character n-gram F-score with word-level n-grams, capturing both fine-grained and coarse-grained translation quality.
- **RTF:** Real-time factor, measuring the speed of the system relative to real-time processing.
- **Latency:** Time taken for the system to process and respond to input.
- TODO : add prosodic metrics for the end-to-end pipeline as well

S2S pipeline: Results

TODO add graphs

Conclusion

- Integrated ASR-MT-TTS systems enable powerful applications
- Continued advances in neural architectures
- Challenges remain in low-resource scenarios
- Future directions: multimodal integration, personalization

Thank You

Thank you for your attention!

Questions?

Contact: your.email@example.com